

PACKAGE INSERT – PAMECIL

1. NAME OF THE MEDICINAL PRODUCT

PAMECIL 500 mg powder for solution for injection/infusion

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each vial contains 500 mg ampicillin as ampicillin sodium.

Excipient(s) with known effect: sodium.

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Powder for solution for injection/infusion.

White or almost white powder.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

PAMECIL is indicated in the treatment of infections caused by ampicillin-sensitive organisms (see section 4.4 and 5.1).

- Soft-tissue infections
- Gastrointestinal infections
- Urinary tract infections
- Bacterial meningitis
- Respiratory tract infections
- Ear, nose and throat infections
- Gonorrhoea
- Urethritis
- Septicaemia
- Typhoid fever
- Peritonitis
- Endocarditis

4.2 Posology and method of administration

PAMECIL is effective in a wide range of dosage regimens. The recommendations given here are commonly applied.

General Recommendations

Route	Dosage	Administration
Adults IM injections	250-500 mg every 6 hours	Dissolve: 250mg in 1.0ml or 500mg in 1.8ml or 1g in 3.5ml water for injections
IV injections/infusion	500mg-1g every 4-6 hours	Dissolve: 250mg or 500mg in 5ml or 1g in 7.4ml water for injections. The resulting solution should be administered slowly over a 3-5 minute period for each 250 to 500 mg dose or over a 10-15 minute period for 1 gram dose. PAMECIL may also be added to infusion fluids or injected, suitably diluted, into the drip tube over a period of 3-4 minutes.
Children IM or IV injection	Up to 2 years: Adult dose 2 – 10 years. Adult dose based on body weight: 25-50mg/kg/day	For IM injection 100mg may be dissolved in 0.5ml water for injections, or in 5ml for IV administration

NOTE:

Solutions should be administered immediately after reconstitution

Respiratory tract and soft tissue infections	Patients weighing 40kg or more: 250-500mg every 6 hours. Patients weighing less than 40kg: 25-50mg/kg/day in equally divided doses at 6 to 8 hour intervals
Gastrointestinal and genitourinary tract infections	Patients weighing 40kg or more: 500mg every 6 hours. Patients weighing less than 40kg: 50mg/kg/day in equally divided doses at 6 to 8 hour intervals
Urethritis (due to <i>N.gonorrhoeae</i>)	Two doses of 500mg each at an interval of 8 to 12 hours. Treatment should be repeated if necessary or extended if required
Bacterial meningitis	Adults and children: 150 to 200mg/kg/day in equally divided doses every 3 to 4 hours (treatment may be initiated with intravenous drip therapy and continued with intramuscular injections)
Septicaemia	Adults and children: 150 to 200mg/kg/day start with intravenous administration for at least 3 days and continue with intramuscular route every 3 to 4 hours
Stability and compatibility	After reconstitution for intramuscular or direct intravenous use, solutions retain their potency for 1 hour. After reconstitution for intravenous infusion solutions in concentrations up to 30mg/ml retain their potency for 2 to 8 hours at room temperature. PAMECIL should not be mixed with blood products or proteinaceous fluids such as protein hydrolysates nor with lipid emulsions.

Duration of use

The duration of use depends on the course of the disease. As a general rule, ampicillin is used for 7 to 10 days, but for at least another 2 to 3 days after the signs of disease have subsided.

For the treatment of infections with beta-hemolytic streptococci, for safety reasons it is recommended to extend the treatment to at least 10 days to prevent late complications (e.g. rheumatic fever, glomerulonephritis).

Method of administration

For instructions on dilution of the product before administration, see section 6.6.

4.3 Contraindications

Hypersensitivity to the active substance, penicillins, history of a severe immediate hypersensitivity (e.g. anaphylaxis) to other beta-lactam antibiotics (e.g. a cephalosporin, carbapenem or monobactam) or to any of the excipients listed in section 6.1.

History of jaundice/hepatic impairment due to ampicillin.

4.4 Special warnings and precautions for use

Before initiating therapy with ampicillin, careful enquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins or other beta-lactam antibiotics.

Ampicillin has a limited spectrum of antibacterial activity. It is not suitable for use as a single agent for the treatment of some types of infections unless the pathogen is already documented and known to be susceptible or there is a very high suspicion that the most likely pathogen(s) would be suitable for treatment with ampicillin. This particularly applies when considering the treatment of patients with intra-abdominal infections, female genital infections and endocarditis.

Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with PAMECIL, careful inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, carbapenems or other beta-lactam agents. If an allergic reaction occurs, PAMECIL must be discontinued immediately and appropriate alternative therapy instituted.

Ampicillin should be avoided if infectious mononucleosis is suspected or the patient suffers from cytomegalovirus

infection or Lymphoid leukaemia since the occurrence of a morbilliform rash has been associated with these conditions following the administration of ampicillin.

Periodic assessment of organ system functions, including renal, hepatic and haematopoietic function is advisable during prolonged therapy.

Ampicillin should be used in the treatment of cystitis only when susceptibility is documented.

Concomitant use of allopurinol during treatment with ampicillin can increase the likelihood of allergic skin reactions.

Prolonged use may occasionally result in overgrowth of non-susceptible organisms.

Prolongation of prothrombin time has been reported rarely in patients receiving ampicillin. Appropriate monitoring should be undertaken when anticoagulants are prescribed concomitantly. Adjustments in the dose of oral anticoagulants may be necessary to maintain the desired level of anticoagulation (see section 4.5 and 4.8).

In patients with renal impairment, the dose should be adjusted according to the degree of impairment (see section 4.2)

During treatment with ampicillin, enzymatic glucose oxidase methods should be used whenever testing for the presence of glucose in urine because false positive results may occur with non-enzymatic methods.

Administration of ampicillin may reduce the efficacy of oral contraceptives.

Supplemental non-hormonal contraceptive measures may be required.

Antibiotic-associated colitis has been reported with nearly all antibacterial agents including ampicillin and may range in severity from mild to life threatening (see section 4.8). Therefore, it is important to consider this diagnosis in patients who present with diarrhea during or subsequent to the administration of any antibiotics. Should antibiotic-associated colitis occur, ampicillin should immediately be discontinued, a physician be consulted and an appropriate therapy initiated. Anti-peristaltic drugs are contra-indicated in this situation.

PAMECIL contains sodium.

PAMECIL 500 mg contains 33.01 mg of sodium per dose. To be taken into consideration by patients on a controlled sodium diet.

4.5 Interaction with other medicinal products and other forms of interaction

Anticoagulants

Co-administration of anticoagulants of the coumarin type can increase the tendency to bleeding.

Digoxin

An increase in the absorption of co-administered digoxin is possible during ampicillin therapy.

Methotrexate

Ampicillin can inhibit the excretion of methotrexate and thereby intensify undesirable effects of methotrexate. The methotrexate levels in the blood should be monitored.

Oral contraceptives

In rare cases the efficacy of hormonal contraceptives may be impaired during ampicillin therapy. The additional use of non-hormonal methods of contraception is therefore recommended.

Probenecid

The co-administration of probenecid inhibits the tubular secretion of ampicillin and leads to higher and longer persisting ampicillin concentrations in serum and bile.

Allopurinol

The simultaneous use of allopurinol during treatment with ampicillin can promote the development of allergic skin reactions.

4.6 Fertility, pregnancy and lactation

Pregnancy

Data from a limited number of exposed pregnancies do not suggest that ampicillin has any adverse effects on pregnancy or the health of the foetus/newborn. No other relevant epidemiological data are available to date. Animal studies have produced no evidence of direct or indirect harmful effects on pregnancy, embryonic/foetal development, childbirth or postnatal development (see section 5.3). Use of ampicillin during pregnancy should be approached with caution.

Breast-feeding

Ampicillin is excreted in breast milk. Breast fed infants may therefore suffer diarrhoea and mucosal yeast colonisation, which in some cases may necessitate the discontinuation of breast feeding. The possibility of sensitisation should be considered. Ampicillin may be used during lactation after a suitable benefit-risk analysis.

Fertility

In animal studies, ampicillin had no effect on fertility (see section 5.3).

4.7 Effects on ability to drive and use machines

No studies on the effects on the ability to drive and use machines have been performed. However, undesirable effects may occur (e.g. allergic reactions, dizziness, convulsions), which may influence the ability to drive and use machines (see section 4.8).

4.8 Undesirable effects

The most commonly reported undesirable effects are skin reactions (pruritis, rash, exanthema, itching), abdominal pain, meteorism, soft stools, diarrhoea, nausea and vomiting.

The undesirable effects derived from clinical studies and post-marketing surveillance, sorted by MedDRA System Organ Class are listed below.

<i>System Organ Class</i>	<i>Very Common</i>	<i>Common</i>	<i>Uncommon</i>	<i>Rare</i>	<i>Very Rare</i>
<i>Infections and infestations</i>			Infection with fungi or resistant bacteria especially during prolonged use		
<i>Blood and lymphatic system disorders</i>					Myelosuppression and haematocytological changes (granulocytopenia, thrombocytopenia, anaemia, pancytopenia); Prolongation of bleeding and prothrombin time ¹ .
<i>Immune system disorders</i> ^{2, 8}	Skin reactions (pruritus, rash, exanthema, itching) ³	Morbilliform rash ⁴ . Exanthem and enanthem in the oral region ⁵ .	More serious allergic reactions such as serum sickness, drug fever, angioneurotic oedema, laryngeal oedema, haemolytic anaemia, allergic vasculitis or nephritis, severe allergic, sometimes		Life-threatening anaphylactic shock ⁶ .

<i>System Organ Class</i>	<i>Very Common</i>	<i>Common</i>	<i>Uncommon</i>	<i>Rare</i>	<i>Very Rare</i>
			vesiculating skin reactions (exfoliative dermatitis, Lyell's syndrome, exudative erythema multiforme, Stevens-Johnson syndrome).		
<i>Nervous system disorders</i> ⁹				Dizziness, headache, myoclonus and seizures (in renal insufficiency or at very high intravenous doses).	
<i>Gastrointestinal disorders</i>	Abdominal pain, nausea, vomiting, meteorism, soft stools, diarrhoea ⁷ .				Pseudomembranous colitis ⁸ (in most cases caused by <i>Clostridium difficile</i>).
<i>Hepatobiliary disorders</i>			Transaminase elevation		
<i>Renal and urinary disorders</i>			Crystalluria on high-dose intravenous administration, acute interstitial nephritis		Acute renal failure with excretion of urine crystals.
<i>General disorders and administration site conditions</i>		Swelling and pain; Localised phlebitis.			

¹See section 4.4

²See sections 4.3 and 4.4

³An immediate-type urticarial reaction generally suggests a true penicillin allergy and necessitates the interruption of treatment and institution of suitable medical measures. Medical advice should be sought regarding the future use of beta-lactam antibiotics.

⁴The typical, measles-like rash develops several (5 to 11) days after the start of treatment.

⁵The incidence of exanthem is higher in patients with infectious mononucleosis or lymphatic leukaemia.

⁶Allergic reactions are more likely to occur in patients with a tendency to allergies.

⁷These undesirable effects are usually mild in nature and frequently subside during, or otherwise after discontinuing the treatment.

⁸If there are signs of pseudomembranous colitis or severe hypersensitivity reactions, the treatment should be discontinued and medical treatment (see section 4.4) provided.

⁹If central nervous excitation, myoclonus or seizures occur, ampicillin should be discontinued and suitable treatment instituted.

4.9 Overdose

Symptoms:

Typical signs of intoxication following the administration of larger amounts of ampicillin have not been observed to date. Long-term therapy is also not associated with specific toxic adverse reactions. The symptoms of over dosage essentially correspond to the profile of undesirable effects (see section 4.8).

The single administration of a larger amount of ampicillin is not acutely poisonous (toxic).

The administration of very high doses can lead to oliguric renal failure and may have effects on nerve cells, for example in the form of central nervous excitation, impairments of muscular function and seizures. The risk of these undesirable effects is increased in patients with severely impaired renal function.

In individual cases, however, these effects were only observed after intravenous administration.

Treatment:

There is no specific antidote in the event of overdose. Treatment comprises symptomatic measures with particular attention to maintaining the water/electrolyte balance.

Ampicillin can be removed from the body by hemodialysis.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Penicillins with extended spectrum, Beta-lactam antibacterials, Penicillins, ATC code J01CA01

Mechanism of action

The mechanism of action of ampicillin is based on inhibition of bacterial wall synthesis (in the growth phase) via blockade of the penicillin-binding proteins (PBPs) such as the transpeptidases. This results in a bactericidal action.

PK/PD relationship

The efficacy depends mainly on the time period for which the active substance level of ampicillin remains above the minimal inhibitory concentration (MIC) of the microorganism.

Mechanisms of resistance

Resistance to ampicillin can be due to the following mechanisms:

- Inactivation by beta-lactamases: ampicillin has only low beta-lactamase stability and is therefore not active against beta-lactamase forming bacteria. Almost all strains of some bacterial species form beta-lactamases. These species are therefore naturally resistant to ampicillin (e.g. *Enterobacter cloacae*, *Klebsiella pneumoniae*).
- Reduced affinity of PBPs for ampicillin: the acquired resistance of pneumococci and other streptococci is due to the modification of existing PBPs as the result of a mutation. Methicillin (oxacillin)-resistant staphylococci, however, are resistant due to the formation of an additional PBP with reduced affinity for ampicillin.
- Insufficient penetration of ampicillin through the outer cell wall of gram-negative bacteria can result in inadequate inhibition of the PBPs.
- Ampicillin can be actively extruded from the cell by efflux pumps.

Partial or complete cross-resistance of ampicillin exists with amoxicillin and to some extent with other penicillins and cephalosporins.

Breakpoints

MIC breakpoints for Ampicillin are those of the European Committee on Antimicrobial Susceptibility Testing (EUCAST).

<i>Microorganism</i>	<i>Susceptible ≤</i>	<i>Resistant ≥</i>
<i>Enterobacteriaceae</i>	8 mg/l	8 mg/l
<i>Enterococcus</i> spp. ¹	4 mg/l	8 mg/l
<i>Haemophilus influenzae</i>	1 mg/l	1 mg/l
<i>Staphylococcus</i> spp. ²	0.12 mg/l	0.12 mg/l
<i>Streptococcus</i> A, B, C G ²	0.25 mg/l	0.25 mg/l
<i>Streptococcus pneumoniae</i>	0.5 mg/l	2 mg/l
Other <i>streptococci</i> ¹	0.5 mg/l	2 mg/l
<i>Neisseria meningitidis</i>	0.12 mg/l	1 mg/l
Gram-negative anaerobes	0.5 mg/l	2 mg/l
Gram-positive anaerobes	4 mg/l	8 mg/l
Non species-specific limit values	2 mg/l	8 mg/l
<i>Listeria monocytogenes</i>	1 mg/l	1 mg/l

¹In endocarditis, refer to national or international endocarditis guidelines for breakpoints.

²Breakpoints values are based on benzylpenicillin breakpoints

Susceptibility

The prevalence of resistance may vary geographically and with time for selected species and local information on resistance is desirable, particularly when treating severe infections. As necessary, expert advice should be sought when the local prevalence of resistance is such that the utility of the agent in at least some types of infection is questionable.

Commonly susceptible species

Aerobic gram-positive microorganisms

Enterococcus faecalis

Staphylococcus aureus (methicillin-sensitive)

Streptococcus agalactiae

Streptococcus pneumoniae (incl. penicillin-intermediate strains)

Streptococcus pyogenes

Streptococci of the Viridans group^

Anaerobic microorganisms

Bacteroides fragilis°

Fusobacterium nucleatum°

Other microorganisms

Gardnerella vaginalis°

Species for which acquired resistance may be a problem

Aerobic gram-positive microorganisms

Enterococcus faecium+

*Staphylococcus aureus**

Staphylococcus epidermidis+

Staphylococcus haemolyticus+

Staphylococcus hominis+

Aerobic gram-negative microorganisms

Escherichia coli

Haemophilus influenzae

Klebsiella oxytoca

Moraxella catarrhalis∞

Neisseria gonorrhoeae

Proteus mirabilis

Proteus vulgaris

Anaerobic microorganisms

Prevotella spp.

Inherently resistant organisms

Aerobic gram-positive microorganisms

Staphylococcus aureus (methicillin-resistant)

Aerobic gram-negative microorganisms

Acinetobacter baumannii

Citrobacter freundii

Enterobacter cloacae

Klebsiella pneumoniae

Morganella morganii

Pseudomonas aeruginosa

Serratia marcescens

Stenotrophomonas maltophilia

Anaerobic microorganisms

Bacteroides spp.

Other microorganisms

Chlamydia spp.

Chlamydophila spp.

Legionella pneumophila

Mycoplasma spp.

Ureaplasma urealyticum

°No current data were available when the table was published. Sensitivity is assumed in the primary literature, standard works and therapy recommendations.

+The resistance rate is above 50% in at least one region.

^Collective name for a heterogeneous group of *Streptococcus* species. Resistance rate can vary depending on the *Streptococcus* species concerned.

∞No recent data available; in studies (older than 5 years) the proportion of resistant strains is reported as ≥10%.

*The resistance rate is <10% in the outpatient setting.

5.2 Pharmacokinetic properties

Distribution

Ampicillin is extensively distributed to tissues, crosses the placental barrier and diffuses into breast milk. Only 5% of the ampicillin concentration in plasma diffuses into cerebrospinal fluid (CSF) with intact meninges. With inflamed meninges, the ampicillin concentration in CSF can increase to 50% of the ampicillin concentration in plasma.

The serum protein binding is 17-20%. The apparent volume of distribution is about 15 L.

Serum level

After oral administration of 1000 mg ampicillin, peak plasma levels of about 5 mg/l are reached after 90 to 120 min. After intramuscular injection, peak plasma levels are reached after 30 to 60 min.

Metabolism

Ampicillin is partly metabolised to microbiologically inactive penicilloates.

Elimination

Ampicillin is eliminated intact mainly by the renal route, but also through bile and faeces. After oral administration, about 40% of a dose is recovered unchanged in the urine. After parenteral administration, about $73 \pm 10\%$ of an administered dose is excreted as unchanged substance in the 0- to 12-hour urine. Up to 10% of a dose is eliminated in the form of biotransformation products. The elimination half-life is about 50 to 60 min. In oliguria, the half-life may be prolonged to 8 to 20 hours. The half-life is also prolonged in newborns (2 to 4 hours). The renal clearance of ampicillin is about 194 ml/min after intravenous administration.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

None.

6.2 Incompatibilities

Ampicillin solutions should always be administered separately, unless compatibility with other infusion solutions or medicines has been established.

This medicinal product must not be mixed with other solutions except those mentioned in section 6.6.

Ampicillin solutions should not be mixed with aminoglycosides, metronidazole and injectable tetracycline derivatives such as oxytetracycline, rolitetracycline and doxycycline. Visual signs of incompatibility are precipitation, clouding and discoloration.

6.3 Shelf-life

3 years.

Use immediately after reconstitution.

6.4 Special precautions for storage

Store below 25°C, in the original packaging.

From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user and would normally not be longer than 24 hours at 2-8°C, unless dilution has taken place in controlled and validated aseptic conditions.

6.5 Nature and contents of container

Clear glass (type III) vials sealed with grey chlorobutyl rubber stopper, aluminium caps with a plastic flip-off seal. Packs of 10, 50, or 100 vials are available.

Not all pack sizes may be marketed.

6.6 Special precautions for disposal and other handling

Reconstitution of PAMECIL should be performed under suitable conditions and with aseptic precautions.

For intramuscular administration: add 1.8 ml of water for injections into the 500 mg vials respectively and shake well.

For intravenous administration: add 5 ml of water for injection into 500 mg vials and shake well. Inject the solution slowly (over a period of 3 to 4 minutes). PAMECIL may be added into compatible infusion fluids. It can also be injected –after suitable dilution- into the drip tube over a period of 3 to 4 minutes.

PAMECIL vials are for single use and any remaining liquid should be discarded following reconstitution and first usage.

7. PRODUCT REGISTRATION HOLDER

KOMEDIC SDN BHD, No. 4 Jalan PJS 11/14, Bandar Sunway, 46150 Petaling Jaya Selangor, MALAYSIA

8. MANUFACTURER

Medochemie (Far East) Ltd., (Aseptic Penicillin facility), No. 10, 12 and 16 Street 27, Vietnam Singapore Industrial Park II A, Vinh Tan Commune, Tan Uyen, Vietnam

9. PRODUCT REGISTRATION NUMBERS

MAL19910079AZ

10. DATE OF REVISION OF THE TEXT

05/2025